Radical equations

1

a $x = 13$ **b** $h = 25$ **c** $x = \frac{9}{64}$ **d** $k = 16$

2 A radical equation is an equation where the variable is inside a radical.

3 No

4 No

5 Yes

6

a $y = 49$ **b** $x = 25$ **c** $p = 9$ **d** $y = \frac{9}{25}$

e $x = \frac{49}{144}$ **f** $x = 64$ **g** $y = 4$ **h** $c = 9$

i $b = 25$ **j** $a = 9$ **k** $b = \frac{1}{25}$ **l** $a = 4$

7

a $x = 30$ **b** $x = 5$ **c** $x = 54$ **d** $x = 11, -3$

e $x = 4, 2$ **f** $x = 18$ **g** $x = 16$ **h** $x = 22$

i $x = 135$ **j** $x = 121$ **k** $x = 4, -4$

8

a No

b Ray did not know that $\sqrt{x+3}$ must always be non-negative. So at step 2, he should have known there was no solution.

9

a Tom did not divide every term by 2 in step 2, so he should have also divided -10 by 2.

b In step 4 Katrina took the square root of 9, when she should have squred it.

c $x = 10$

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10 Example answers:

- In order for the left-hand side to equal zero, the square root of the number would have to be negative.
- The left-hand side is greater than zero for all values of x.

11

a $x = \frac{1}{4}, \frac{1}{25}$ **b** $x = \frac{1}{25}$

12

a $x = 9$ **b** No

13 Raise each side of the equation to the power $\frac{5}{3}$.

14

a $x = 57$ **b** $x = 4$ **c** $x = 79$

15

a $x = 9$ **b** $x = \pm 128$ **c** $x = 81$ **d** $x = \pm 128$

Applications

16

a

V	40	400	4 000
b	3.2	10.0	31.6

b No

17

a $t = \sqrt{4 - x} + 4$ **b** $t = 6$ **c** 0.2 seconds

18

a 8 **b** Fresh gale **c** $s = 115$ **d** $55 \leq s \leq 64$

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